

Elevate Labs SQL Developer Internship

Task 4 - SQL Joins (Inner, Left, Right, Full).

1. Schema / Sample tables –

-- For example, in MySQL or SQLite:

```
CREATE TABLE Customers (  
  customer_id INT PRIMARY KEY,  
  name        VARCHAR(100),  
  city        VARCHAR(100)  
);
```

```
CREATE TABLE Orders (  
  order_id    INT PRIMARY KEY,  
  customer_id INT,          -- foreign key to Customers  
  order_date  DATE,  
  amount      DECIMAL(10,2)  
);
```

2. Insert some sample data –

```
INSERT INTO Customers (customer_id, name, city) VALUES  
(1, 'Alice', 'Mumbai'),  
(2, 'Bob', 'Delhi'),  
(3, 'Charlie', 'Pune'),  
(4, 'Diana', 'Nagpur');
```

```
INSERT INTO Orders (order_id, customer_id, order_date, amount) VALUES  
(101, 1, '2025-09-01', 250.00),  
(102, 1, '2025-09-10', 450.00),  
(103, 2, '2025-09-15', 300.00),  
(104, 5, '2025-09-20', 500.00);
```

- **INNER JOIN –**

```
SELECT  
  c.customer_id,  
  c.name,  
  o.order_id,  
  o.order_date,  
  o.amount  
FROM Customers c  
INNER JOIN Orders o
```

ON c.customer_id = o.customer_id;

- **LEFT (OUTER) JOIN –**

```
SELECT
    c.customer_id,
    c.name,
    o.order_id,
    o.order_date,
    o.amount
FROM Customers c
LEFT JOIN Orders o
    ON c.customer_id = o.customer_id;
```

- **RIGHT (OUTER) JOIN –**

```
SELECT
    c.customer_id,
    c.name,
    o.order_id,
    o.customer_id AS o_cust_id,
    o.order_date,
    o.amount
FROM Customers c
RIGHT JOIN Orders o
    ON c.customer_id = o.customer_id;
```

- **FULL (OUTER) JOIN –**

```
SELECT
    c.customer_id,
    c.name,
    o.order_id,
    o.customer_id AS o_cust_id,
    o.order_date,
    o.amount
FROM Customers c
FULL OUTER JOIN Orders o
    ON c.customer_id = o.customer_id;
```